



Breast Cancer Wisconsin (Diagnostic)

Donated on 10/31/1995

Diagnostic Wisconsin Breast Cancer Database.

Dataset Characteristics

Multivariate

Subject Area

Health and Medicine

Associated Tasks

Classification

Feature Type

Real

Instances

569

Features

30



DOWNLOAD (50.1 KB)



IMPORT IN PYTHON

CITE

37 citations

482926 views

Keywords

health

cancer

Dataset Information

Additional Information

Features are computed from a digitized image of a fine needle aspirate (FNA) of a breast mass. They describe characteristics of the cell nuclei present in the image. A few of the images can be found at <http://www.cs.wisc.edu/~street/images/>

Separating plane described above was obtained using Multisurface Method-Tree (MSM-T) [K. P. Bennett, "Decision Tree Construction Via Linear Programming." Proceedings of the 4th Midwest Artificial Intelligence and Cognitive Science Society, pp. 97-101, 1992], a classification method which uses linear programming to construct a decision tree. Relevant features were selected using an exhaustive search in

Creators



William Wolberg



Olvi Mangasarian



Nick Street



W. Street

DOI

the space of 1-4 features and 1-3 separating planes.

The actual linear program used to obtain the separating plane in the 3-dimensional space is that described in: [K. P. Bennett and O. L. Mangasarian: "Robust Linear Programming Discrimination of Two Linearly Inseparable Sets", Optimization Methods and Software 1, 1992, 23-34].

This database is also available through the UW CS ftp server:

```
ftp ftp.cs.wisc.edu
cd math-prog/cpo-dataset/machine-learn/WDBC/
```

SHOW LESS ^

Has Missing Values?

No

10.24432/C5DW2B

License

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This allows for the sharing and adaptation of the datasets for any purpose, provided that the appropriate credit is given.

Introductory Paper ^

[Nuclear feature extraction for breast tumor diagnosis](#)

By W. Street, W. Wolberg, O. Mangasarian. 1993
Published in Electronic imaging

Variables Table ^

Variable Name	Role	Type	Description	Units	Missing Values
ID	ID	Categorical			no
Diagnosis	Target	Categorical			no

Variable Name	Role	Type	Description	Units	Missing Values
radius1	Feature	Continuous			no
texture1	Feature	Continuous			no
perimeter1	Feature	Continuous			no
area1	Feature	Continuous			no
smoothness1	Feature	Continuous			no
compactness1	Feature	Continuous			no
concavity1	Feature	Continuous			no
concave_points1	Feature	Continuous			no
symmetry1	Feature	Continuous			no
fractal_dimension1	Feature	Continuous			no
radius2	Feature	Continuous			no
texture2	Feature	Continuous			no
perimeter2	Feature	Continuous			no
area2	Feature	Continuous			no
smoothness2	Feature	Continuous			no

Variable Name	Role	Type	Description	Units	Missing Values
compactness2	Feature	Continuous			no
concavity2	Feature	Continuous			no
concave_points2	Feature	Continuous			no
symmetry2	Feature	Continuous			no
fractal_dimension2	Feature	Continuous			no
radius3	Feature	Continuous			no
texture3	Feature	Continuous			no
perimeter3	Feature	Continuous			no

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Additional Variable Information

- 1) ID number
- 2) Diagnosis (M = malignant, B = benign)
- 3-32)

Ten real-valued features are computed for each cell nucleus:

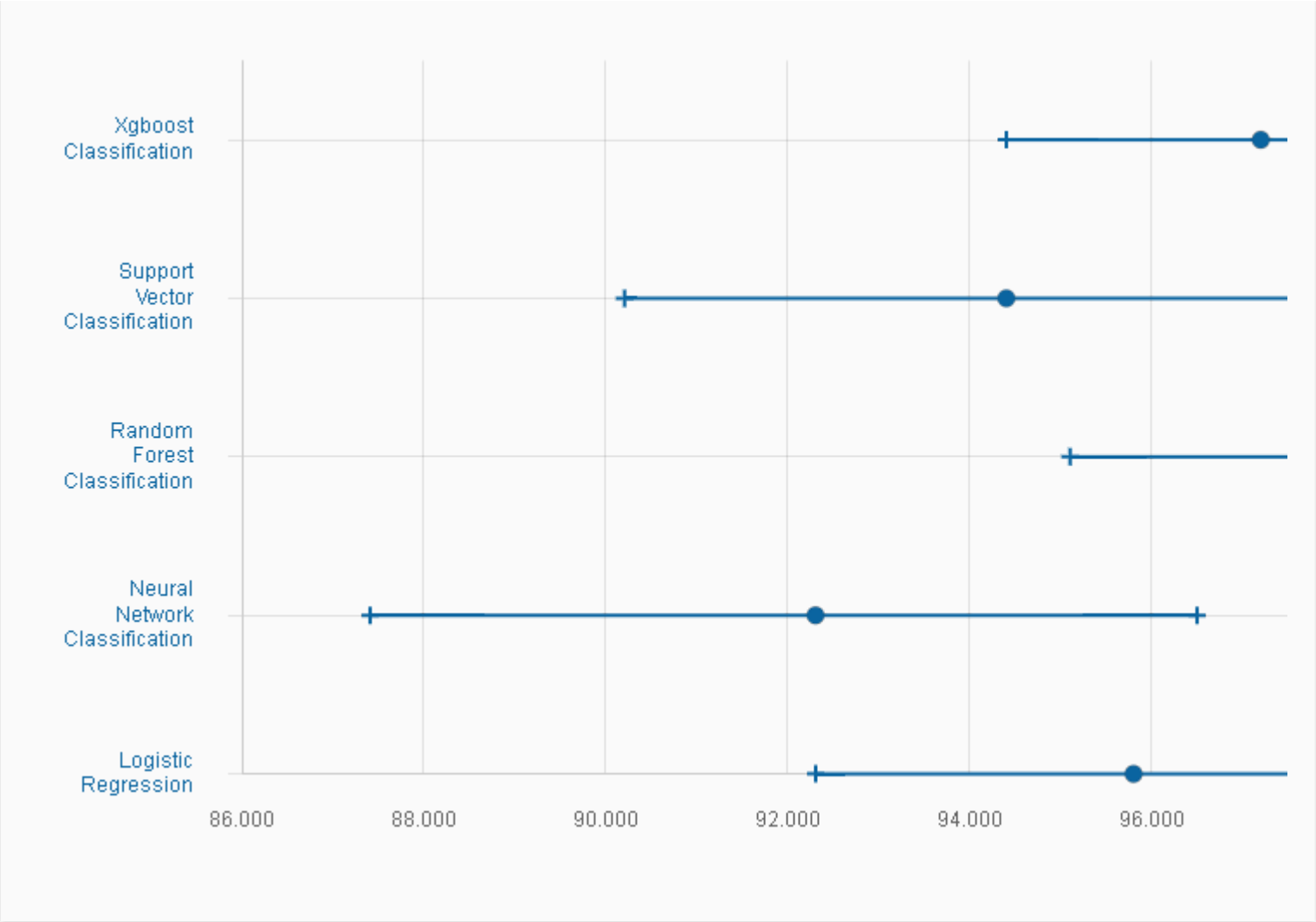
- a) radius (mean of distances from center to points on the perimeter)
- b) texture (standard deviation of gray-scale values)
- c) perimeter
- d) area
- e) smoothness (local variation in radius lengths)
- f) compactness ($\text{perimeter}^2 / \text{area} - 1.0$)
- g) concavity (severity of concave portions of the contour)
- h) concave points (number of concave portions of the contour)
- i) symmetry
- j) fractal dimension ("coastline approximation" - 1)

SHOW LESS ^

Baseline Model Performance ^

Accuracy

Precision



Dataset Files



File	Size
wdbc.data	121.2 KB

File	Size
wdbc.names	4.6 KB

Papers Citing this Dataset



SORT BY YEAR, DESC

[Machine learning in medicine: a practical introduction](#)

By Jenni Sidey-Gibbons, Chris Sidey-Gibbons. 2019
Published in BMC medical research methodology.

[Data Augmentation Using GANs](#)

By Fabio Tanaka, Claus Aranha. 2019
Published in ArXiv.

[Empirical study towards understanding line search approximations for training neural networks](#)

By Younghwan Chae, Daniel Wilke. 2019
Published in

[Interpretable Counterfactual Explanations Guided by Prototypes](#)

By Arnaud Looveren, Janis Klaise. 2019
Published in ArXiv.

[PIDT: A Novel Decision Tree Algorithm Based on Parameterised Impurities and Statistical Pruning...](#)

By Daniel Stamate, Wajdi Alghamdi, Daniel Stahl, Doina Logofatu, Alexander Zamyatin. 2018
Published in AIAI.

Reviews

3.8



(2 ratings)

LOGIN TO WRITE A REVIEW

Ümit Arslan

 -

2/23/2025

Ramya Ramana

 -

11/15/2024



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